

POWER OASIS™ MEDICAL

Isolated power that moves with your OR.

The first portable power station designed to NFPA 99 Category 1 isolated power requirements — hospital-grade isolation transformer, line isolation monitoring, and seamless UPS transfer in a single cart.

- **NFPA 99 Category 1** isolated power architecture[4]
- **IS1800HG** — UL 60601-1 listed isolation transformer
- **Line Isolation Monitor** — alarm at ≤5mA hazard current
- **<20 ms UPS** — OR equipment never reboots[1]
- **3h 34m @ 1,000W** — full coverage for routine OR cases

4,800	3,571	1,800	<20	3h 34m	~214
Wh	Wh usable AC	W	ms[1]	@ 1,000W	lbs[2]
GROSS ENERGY	USABLE OUTPUT	ISOLATED OUTPUT	UPS TRANSFER	OR RUNTIME	CART WEIGHT

OR RUNTIME AT LOAD

Coverage for the clinical scenarios that matter

OR Load	Runtime	Clinical Context
500W	7h 08m	Anesthesia monitor · infusion pump · lighting
750W	4h 45m	Multi-device light OR load
1,000W	3h 34m	Typical mixed OR — anesthesia + monitor + ESU
1,500W	2h 22m	Heavy OR load — near IS1800HG capacity
1,800W	1h 59m	IS1800HG rated maximum

At 80% DoD · 93% inverter efficiency · 100Ah battery. Lightweight 60Ah configuration available (~30 lbs · ~$900 · 2h 08m @ 1,000W). Designed as backup for unexpected outages — recharges in ~2.9 hrs.

KEY SPECIFICATIONS

Energy	**4,800 / 3,571**	Wh	*Gross / usable AC*
Inverter	**48/3000**		*Online UPS — never interrupts*
Isolation transformer	**IS1800HG · 1,800W**		*Tripp Lite · UL 60601-1 3rd ed listed*
Output leakage	**<100**	µA	*Faraday shield · NFPA 99 limit*
Line isolation monitor	**≤5mA alarm**		*Visual + audible · front panel[5]*
Battery	**48V 100Ah LiFePO4**		*~75 lbs[2] · 60Ah lightweight option*

Recharge	**~2.9 hrs**		*35A integrated charger*
Outlets	**8 positions (4 duplex)**		*Red hospital-grade NEMA 5-20R-HG[3]*
AC input	**15A limit · HG plug**		*IS1800HG thermal rating*
EMI filter	**FN9246B medical**		*<2µA leakage — B-type*
Enclosure	**6061-T6 aluminum**		*26×22×34 in · wipe-clean*
Weight	**~214 lbs[2]**		*Verify facility placement policy*

[1] Per manufacturer specification with UPS Function enabled; transfer performance verified by oscilloscope during factory acceptance. [2] Estimated — pending vendor confirmation. [3] Render shows 6-outlet design study; production spec is 4 duplex (8 positions). [4] Compliance path under regulatory review — see notices. [5] LIM make/model selection in progress.

ISOLATED POWER ARCHITECTURE — NFPA 99 CATEGORY 1

Why each element exists and what it protects

Element	Component	Function
Isolation transformer	Tripp Lite IS1800HG	Breaks ground reference — first fault does not interrupt power or endanger patient
Line isolation monitor	LIM (model TBD[5]) · front panel	Monitors isolation integrity · alarms ≤5mA · powered from IS1800HG output side per NFPA 99
Output leakage	<100µA verified	IS1800HG Faraday shield — measured on each production unit
No GFCI on OR outlets	Intentional design	GFCI would cut ALL OR power on first fault — defeats isolated power purpose
Seamless UPS	Online inverter/charger	Inverter runs continuously — OR equipment never sees the transfer
Fuse monitoring	DC distribution + CAN bus	Blown DC fuse → visible system display alarm — no silent failures in the OR

ADDITIONAL SPECIFICATIONS

Complementing key specifications on page 1

Rated output	**3,000 VA · 6,000 W surge**		*inverter/charger*
Max discharge	**0.40C**		*At 1,800W · 60% headroom to 1C*
Charge voltages	**58.4 / 54.4**	V DC	*LiFePO4 absorption / float*
Efficiency	**~93% inverter · ~97% IS1800HG**		*Combined ~90%*

IS1800HG weight	**58**	lbs	*Two-person install lift*
Cooling	**2× 120mm fans**		*GX relay · locking casters*
Input plug	**NEMA 5-20P-HG**		*Hospital-grade*
Operating temp	**−20 to +40**	°C	*Hospital environment*

WARRANTY & SUPPORT

Coverage	Details
Product warranty	2 years parts/labor (target) · 5 years battery and power electronics
Annual service	LIM verification · leakage measurement · battery health — included year 1
Remote diagnostics	Cloud monitoring — biomedical team and our support see identical live data
Compliance docs	Complete NFPA 99 documentation package ships with each unit

⚠ IMPORTANT NOTICES

- CONCEPT STAGE — NOT for clinical use. NFPA 99 compliance path under regulatory review; deployment pending confirmation.
- EMI filter is FN9246B medical B-type exclusively — standard filters violate leakage limits.
- No GFCI on OR outlets is intentional — see architecture above. Render shows 6-outlet design study; production is 8 positions.

Planning an OR renovation or resilience program?

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